

Parking Space Detection - How to Run the Project

CAI2840C | Step-by-step guide

Quick start (demo - no large downloads)

1. Unzip the file

Extract the project ZIP into a writable folder. A local path (e.g. C:\Parking) is faster than a network share.

2. Open PowerShell in that folder

In File Explorer, open the extracted Parking folder, then use File > Open Windows PowerShell (or right-click > Open in Terminal).

3. Check Python (3.10-3.12 only - not 3.14)

TensorFlow currently has no official wheels for Python 3.14. Prefer 3.12:

```
py -3.12 --version
```

4. Create the virtual environment

```
py -3.12 -m venv .venv
```

5. Install dependencies

```
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

6. Run the demo

Creates synthetic data, trains, evaluates, and writes results under outputs/:

```
.\.venv\Scripts\python.exe run_pipeline.py --demo --epochs 3 --models baseline,mobilenetv3
```

7. Check the results

When you see Pipeline finished, inspect these folders:

```
outputs\metrics\
outputs\figures\
outputs\models\
```

Optional: Jupyter notebook

For a guided walkthrough after installing dependencies:

```
.\.venv\Scripts\python.exe -m jupyter notebook notebooks\01_parking_occupancy_experiment.ipynb
```

Full study (optional)

For the complete experiment, download PKLot and CNRPark-EXT into data/raw/ (see README for expected folder layout), then run:

```
.\.venv\Scripts\python.exe -m src.prepare_data
.\.venv\Scripts\python.exe -m src.train --model all
.\.venv\Scripts\python.exe -m src.evaluate --model all
```

Troubleshooting

Problem	What to do
py -3.12 not found	Install Python 3.12 from python.org (add to PATH). Do not use 3.14.
pip / TensorFlow fails	Use the venv python shown above. Upgrade pip, then reinstall requirements.txt.
ModuleNotFoundError	Re-run pip install from the project root with .\venv\Scripts\python.exe.
Slow or UNC path issues	Copy the project to a local folder (e.g. C:\Parking) and run from there.
Demo too slow	Keep --epochs 3 and models baseline,mobilenetv3 as in step 6.
No outputs created	Wait for Pipeline finished, then open outputs\metrics and outputs\figures.

Tip: Always run commands with .\venv\Scripts\python.exe so packages install into the project environment.